

IMPACT ANALYSIS

CR06: Dashboard Sidebar & Course Management

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Department of Interdisciplinary Studies
Information and Communication Technology
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Change Request ID	CR06
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Change Title	Dashboard Sidebar & Course Management

1. Purpose of the Impact Analysis

This impact analysis identifies the parts of the system that may be affected by CR06. The analysis is based on the supplied implementation plan, which requires a Dashboard Sidebar, nested dashboard routing, separation of existing student-management logic, a new Courses management interface, course API services, backend course routes/controllers, and a courses database table.

2. Overall Impact Assessment

Overall impact: HIGH. CR06 is a multi-layer change affecting frontend routing and layout, existing student-management code, new course-management screens, backend API functionality, and the database schema. The main regression risks are the student-view refactor, nested routing, and frontend-to-backend course integration.

3. Detailed Impact Analysis

Impact Area	Affected Components / Files	Impact	Potential Effect	Recommended Verification
Dashboard Routing	frontend/src/routes/AppRoutes.jsx; frontend/src/pages/DashboardPage.jsx	High	Incorrect nested route configuration may prevent access to Students or Courses.	Verify /dashboard/students is the default module and /dashboard/courses opens Courses.
Dashboard Layout	DashboardPage.jsx; DashboardSidebar.jsx; dashboard-sidebar.css	High	The layout refactor may alter existing dashboard structure, navigation, or visual behaviour.	Verify sidebar placement, active states, header, and module outlet together.
Student Management	DashboardPage.jsx; StudentsView.jsx	High	Moving student fetching, search, and table logic can cause functional regressions.	Test student loading, search, table display, and navigation after refactoring.
Courses Frontend	CoursesView.jsx; CourseFormModal.jsx; api.js/courseService.js	High	Incomplete UI or API integration may prevent creating or displaying courses.	Test Add Course, form fields, validation, create request, and course list rendering.
Backend API	courseRoutes.js; courseController.js; app.js	High	Route, controller, or registration errors can break course operations.	Test POST /api/courses and GET /api/courses directly and through the frontend.

Database	database/schema.sql	High	An incorrect or incomplete courses table can prevent data persistence or retrieval.	Verify schema creation and successful course insert/retrieval.
Access / Security	Existing Admin Dashboard access model; new course endpoints	Medium	The plan identifies Courses as an administrator feature but does not specify endpoint authorization details.	Preserve the existing Admin Dashboard access model and document any endpoint authorization used.
Integration / Regression	Students, Courses, routing, API, database	High	Because several layers change together, integration defects may occur.	Perform end-to-end verification from dashboard navigation through course persistence and retrieval.
Additional File Changes	Any files beyond the identified components	Medium	Unplanned changes can expand scope and make the implementation difficult to trace.	Document and justify all additional modified files in the Implementation Report.

4. Key Risks and Mitigation

- Routing risk: verify nested routes and ensure /dashboard/students remains the default module.
- Regression risk: compare existing Students functionality before and after moving the logic to StudentsView.jsx.
- Integration risk: verify the complete flow from CourseFormModal.jsx to POST /api/courses, database storage, GET /api/courses, and CoursesView.jsx display.
- Schema risk: verify the courses table before frontend integration testing.
- Access-control risk: retain the existing administrator access model and explicitly document any new backend authorization behaviour.

5. Impact Conclusion

CR06 has a high overall implementation impact because it modifies the dashboard structure and introduces a new management module across the frontend, backend, and database layers. The change is manageable provided that routing, student-management regression, course API integration, and database persistence are verified systematically. Any additional files modified beyond those identified in the implementation plan should be documented and justified in the Implementation Report.